## 0.1 Topological Maps

**Definition 0.1.1.** Let $X, Y$ be Topological Spaces. A map $f : X \to Y$ is said to be *continuous* at $x_0 \in X$ if:

For any open $V \in \mathcal{T}_Y$ with $f(x_0) \in V$, there is an open $U \in \mathcal{T}_X$ with $x_0 \in U$ such that $f(U) \subset V$

If $f$ is continuous at every point in the domain, then $f$ is called a *continuous map*.

**Definition 0.1.2.** Let $X, Y$ be Topological Spaces.
A map $f : X \to Y$ is called an *open map* if: For any open $U \subseteq X$, $f[U] \subseteq Y$ is open in $Y$.
A map $f : X \to Y$ is called a *closed map* if: For any closed $C \subseteq X$, $f[C] \subseteq Y$ is closed in $Y$.

**Theorem 0.1.3.** Let $f : X \to Y$ be a map. Then, The Followings are Equivalent:

**a)** $f$ is Continuous Map.

**b)** For any closed $C \subset Y$, $f^{-1}[C] \subset X$ be closed.

**c)** For any subset $A \subset X$, $f[\overline{A}] \subset \overline{f[A]}$.

**d)** For any subset $B \subset Y$, $f^{-1}[A^\circ] \subset (f^{-1}[B])^\circ$.

*Proof.*
**a)** $\implies$ **b)** Let $C \subset Y$ is closed. Then,

$$f^{-1}[Y \setminus C] = X \setminus f^{-1}[C]$$

is open in $Y$, thus $f^{-1}[C]$ is closed.
**b)** $\implies$ **c)** Let $A \subset X$ be given. Then,

$$A \subset f^{-1}[f[A]] \subset \underbrace{f^{-1}[\overline{f[A]}]}_{\text{closed by } b)} \implies \overline{A} \subset f^{-1}[\overline{f[A]}] \implies f[\overline{A}] \subset f[f^{-1}[\overline{f[A]}]] \subset \overline{f[A]}$$

**c)** $\implies$ **d)** Let $B \subset Y$, put $A = f^{-1}[Y \setminus B] = X \setminus f^{-1}[B]$. Then,

$$f[\overline{A}] = f[\overline{X \setminus f^{-1}[B]}] = f[X \setminus (f^{-1}[B])^\circ] \text{ and } \overline{f[A]} = \overline{f[f^{-1}[Y \setminus B]]} \subset \overline{Y \setminus B} = Y \setminus B^\circ$$

By c),

$$\begin{aligned} & f[\overline{A}] = f[X \setminus (f^{-1}[B])^\circ] \subset \overline{f[A]} \subset Y \setminus B^\circ \\ \implies & X \setminus (f^{-1}[B])^\circ \subset f^{-1}[f[X \setminus (f^{-1}[B])^\circ]] \subset f^{-1}[Y \setminus B^\circ] = X \setminus f^{-1}[B^\circ] \\ \implies & f^{-1}[B^\circ] \subset (f^{-1}[B])^\circ \end{aligned}$$

**d)** $\implies$ **a)** Let $U \subset Y$ be an open set. By d),

$$f^{-1}[U] \overset{U \text{ open}}{=} f^{-1}[U^\circ] \subset (f^{-1}[U])^\circ$$

Meanwhile, reverse inclusion is clear, thus $f^{-1}[U] = (f^{-1}[U])^\circ$, open. □

**Lemma 0.1.4.** Let $X, Y$ be Topological Space. Then,

1. $f : X \to Y$ is open map **if and only if** For any $A \subset X$, $f[A^\circ] \subset (f[A])^\circ$.
2. $f : X \to Y$ is closed map **if and only if** For any $A \subset X$, $\overline{f[A]} \subset f[\overline{A}]$.

**Lemma 0.1.5.** Let $X, Y$ be Topological Space, and $f : X \to Y$ be a bijection. Then, TFAE:

1. $f$ is open map.
2. $f$ is closed map.
3. $f^{-1} : Y \to X$ be continuous map.

Clearly, Homeomorphism is open, closed, continuous map.

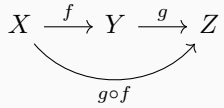
**Lemma 0.1.6.** Let $f : X \to Y$ be a Homeomorphism, $A \subset X$. Then, followings hold:

1. $f[\overline{A}] = \overline{f[A]}$.
2. $f[A^\circ] = (f[A])^\circ$.

*Proof.* 1. is clear by $f$ is continuous, and closed map.
2. $\subset$) $A^\circ \subset A \implies f[A^\circ] \subset f[A] \implies f[A^\circ] \subset (f[A])^\circ \subset f[A]$ by $f$ open map.
2. $\supset$) Let $x \in (f[A])^\circ$ be given. Then, there is an open $\mathcal{U} \in \mathcal{T}$ such that $x \in \mathcal{U} \subset f[A]$.
Now, $f^{-1}[x] \subset f^{-1}[\mathcal{U}] \subset f^{-1}[f[A]] = A$ by $f$ is bijection, this implies
$(f^{-1}[\mathcal{U}])^\circ = f^{-1}[\mathcal{U}] \subset A^\circ \implies f^{-1}[x] \subset f^{-1}[\mathcal{U}] \subset A^\circ \implies f[f^{-1}[x]] = x \in f[A^\circ]$. □

**Theorem 0.1.7.** Let $X, Y, Z$ be Topological Spaces, and $f : X \to Y$, $g : Y \to Z$ be functions.

1. If $f, g$ are Continuous map, then $g \circ f$ is Continuous map.
2. If $f, g$ are Open map, then $g \circ f$ is Open map.
3. If $f, g$ are Closed map, then $g \circ f$ is Closed map.




Above three statements are trivial.

1. If $g \circ f$ is Open map, $f$ is Continuous onto map. Then, $g$ is Open map.
2. If $g \circ f$ is Open map, $g$ is Continuous one-to-one map. Then, $f$ is Open map.

*Proof.* 1) Let $U \in \mathcal{T}_Y$ be an open set. Since $f$ is Continuous map, $f^{-1}[U]$ is open of $X$.
Now,

$$\underbrace{(g \circ f)[\underbrace{f^{-1}[U]}_{\text{open}}]}_{\text{image of open map}} = g[f[f^{-1}[U]]] \overset{\text{onto}}{=} g[U] \in \mathcal{T}_Z$$

2) Let $U \in \mathcal{T}_X$ be an open set. Since $g \circ f$ is Open map, $(g \circ f)[U]$ is open of $Z$.
Now, by $g$ is Continuous one-to-one,

$$g^{-1}[(g \circ f)[U]] = g^{-1}[g[f[U]]] \overset{1 \text{ to } 1}{=} f[U] \in \mathcal{T}_Y$$

□

**Lemma 0.1.8.** *Pasting Lemma*
Suppose that $X, Y$ are Topological Space, and $A, B \subset X$ such that $X = A \cup B$.
If both $A, B$ are Open or Closed, $f : A \to Y$ and $g : B \to Y$ are Continuous map such that $f|_{A \cap B} = g|_{A \cap B}$, then

$$f \cup g : X \to Y : \begin{cases} f(x) & x \in A \\ g(x) & x \in B \end{cases}$$

is Continuous map.